

,Princeton University
Mathematical Intermediate Microeconomics (ECO 310)
Midterm Spring 2023

Instructions: You have 80 minutes to complete this exam, plus 10 minutes to upload your answers. There are 6 multiple choice questions and 3 short answer questions. **In the Multiple Choice section, simply indicate the correct answer. In the Short Answer section, show your work and explain your answers for full credit!**

- The exam is open-book, and calculators are permitted. However, please work **alone**.
- **If you are unsure whether you're interpreting a question correctly, state your interpretation and proceed (or email me if you prefer).**

The honor code applies, as usual.

Q1: Multiple Choice (24 points, 4 each)

1. Jake tells you that when comparing any two numbers a and b , he prefers a if and only if it's at least as large as $b + 1$. Jake is:
 - (a) Rational
 - (b) Irrational because these preferences are incomplete
 - (c) Irrational because these preferences are intransitive
 - (d) Two of these are correct
2. A consumer's preferences over good 1 and good 2 are described by $u(x_1, x_2) = x_1^{\frac{1}{2}}x_2$. If this consumer behaves optimally, he will:
 - (a) Spend $\frac{1}{3}$ of income on good 2.
 - (b) Spend $\frac{1}{2}$ of income on good 2.
 - (c) Spend $\frac{2}{3}$ of income on good 2.
 - (d) None of these is correct.
3. A consumer initially buys 5 oranges and 10 apples. Then an apple tax increases the price per apple by \$0.50, but the consumer is at the same time given a \$5 fruit subsidy. Which of the following best describes this situation?
 - (a) The consumer is most likely better off now (with the tax and subsidy) than he was originally.
 - (b) The consumer is most likely worse off now (with the tax and subsidy) than he was originally.
 - (c) The consumer is definitely indifferent between his original and current situations.
4. A consumer gets utility $u(L, R) = \min\{L, R\}$ from L left shoes and R right shoes. If $m = 10$, $p_L = 1$, and p_R increases from 1 to 4, the change in demand for right shoes due to the substitution effect is:
 - (a) 0
 - (b) -1
 - (c) -2
 - (d) -3
 - (e) None of these is correct
5. At the end of Lecture 10, we saw the following "paradox": many people prefer saving 200 of 600 people for sure, to a risky option with a $\frac{1}{3}$ chance of saving all 600 people and a $\frac{2}{3}$ chance of saving nobody. But they prefer a risky option with a $\frac{1}{3}$ chance of killing nobody and a $\frac{2}{3}$ chance of killing everybody, to an option that kills 400 people for sure.

If people care about the # people alive in the end, then

 - (a) This is inconsistent with EU maximization

- (b) This violates consequentialism
 - (c) This violates independence
 - (d) Two of these are true
 - (e) All of these are true
6. Which of the following would a risk-averse consumer **definitely** prefer to a 50-50 \$0 or \$200 gamble:
- (a) \$90 lump-sum
 - (b) \$110 lump-sum
 - (c) A 50-50 \$50 or \$150 gamble
 - (d) Two of a-c are definitely preferred
 - (e) Three of a-c are definitely preferred

Q2: Proofs and General Concepts (20 Points)

- a. (12) Assume that Melsi and Mavi (my dogs) are equally wealthy. Both are risk-averse, but Mavi is **more** risk-averse. Prove through the following steps that any time Mavi prefers accepting a risky bet (to rejecting it), so does Melsi:
- i. (6) Show that someone prefers accepting (rather than rejecting) a risky bet if and only if their certainty equivalent for the gamble exceeds their initial wealth.
 - ii. (6) If Mavi is more risk-averse than Melsi, then from Lecture 10, her risk premium is higher for every gamble. Prove that this implies that for a particular gamble, Melsi's certainty equivalent is higher than Mavi's.
- b. (18) You are working for a law firm that represents an elderly man who wants to leave his fortune to his dog. His evil family plans to have him committed if he is caught violating WARP. Previously, he consumed $(4, 2)$ when prices were $(1, 3)$. Prices are now $(3, 5)$. Which bundles (x_1, x_2) would get him committed if he purchased them now? (Just write out the inequality(ies) that must hold, no need to simplify).

Q3: Demand (37 Points)

Harry's preferences over netflix hours (N) and cable hours (C) are represented by

$$u(N, C) = C^\alpha + 2CN$$

where $\alpha > 0$ is a parameter.

- a. (8) Show that if $\alpha \leq 2$, then these preferences are convex.
- b. (5) Write the equation for Harry's indifference curve through the point (1,1). (No need to simplify).
- c. (12) Find Harry's (Marshallian) demand function if $\alpha = 2$, assuming income m and prices p_C and p_N per hour of cable and netflix.
- d. (12) Finally, suppose Harry's limited resource is time rather than money: Netflix and cable are both offered free of charge in his dorm room, but he has limited time available to watch them. (i) What is his budget constraint if he has T hours available? (ii) Again assuming that $\alpha = 2$, how much time T does he need to get utility u^* ?

Q4: Expected Utility (31 Points)

Suppose you are *risk-averse*, have \$100 (no wealth besides this), and are considering investing some of your \$100 in an asset with 3 possible return rates: $r_1 = -1$ (which happens with probability $q \leq \frac{1}{2}$), $r_2 = 1$ (which also happens with probability $q \leq \frac{1}{2}$), and $r_3 = 2$ (this happens whenever the other two possible return rates do not). Your other option is to hide the money in a sock drawer. Recall that \$ x invested at return rate r **changes** your wealth by \$ xr .

- a. (12) Write out your expected utility from investing \$ x of your \$100 in this asset, and keeping the remaining \$100- x in a sock drawer. (Hint: consider drawing a probability tree, with one branch for each way the uncertainty can resolve; at the end of each branch figure out your final utility if this happens.)
- b. (14) If you can invest any amount x between \$0 and \$100 (inclusive, decimals allowed),
 - i. Explain: at $q = 0$, you should invest your entire \$100 in the asset.
 - ii. Find all values of q (if any) for which it is optimal to keep all money in your sock drawer.
- c. (5) If your only options were to invest \$20 or \$80, how would you figure out which is better?